

RESPONSE TO THE R44 REFEREE REPORT

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1. OVERVIEW

We thank the referee for separating the incomplete publication object from the substantive mathematical and computational issues. The present revision addresses both. It supplies the full article and ordinary reproducible source; makes the restart-price theorem and its exact gap identity the main result; integrates the price-tree and tie evidence; and adds paired operating-witness theory and a new independent precision audit. The policy class, original continuous-state objective, and unfavorable results are preserved.

The report addressed here is the R44 report of September 25, 2026, at commit `bdfd4e94b74a41ef904158d0061d10a6fcf39c8c`. The computational baseline is the subsequent R46 commit `fb88b1122ce5709d36104993937083b57470d1de`. We do not mistake that later development for a new referee report or for an untouched evaluation. The original frozen B and A counts remain four and five of twelve primary cases. The retrospective price search reaches six primary and three tie targets. Native SCIP lower verification, a difficult learned operating-oracle application, a calibrated empirical application, and numerical closure of the original thirty positive-cost continuous intervals remain uncompleted evidence requests. Each is identified below rather than being marked resolved through a presentation change.

2. MAJOR FINDINGS

2.1. *R44-F1: A complete, normally readable review object*

The revision now supplies a single integrated article, a full numerical supplement, this response, a computation report, and a review index. The ordinary TeX sources, generated tables, proof readers, PDF build records, and content manifest are part of the review object. The build verifies that inherited tracked files are unchanged. The previous R45 draft is preserved as source and as a separate historical reading volume rather than being silently discarded. No referee needs to decode a staging fragment to read the article. The final publication commit, identified in the review index and build metadata, is the authoritative object.

2.2. *R44-F2: Target attainment and nontrivial successes*

The abstract and main results retain the original frozen primary counts: four of twelve Bellman and five of twelve aggregate targets at absolute width 10^{-3} . The subsequent price-strengthened search attains six of twelve primary targets and three of four tie targets; all sixteen cases are shown. I0 is zero cost, Q0/I3/Q3 close at the price root, and M0/Q1 close only after price-tree branching. The new witness audit consists of four precision variants of the same sixteen models, not a prospective benchmark. Thus the evidence is stronger without changing the target or redefining success. A new untouched suite with a majority of difficult positive-cost closures has not been executed; the retrospective extension is labeled accordingly.

2.3. *R44-F3: Formulation reversals and a valid portfolio*

The article no longer uses formulation strength as a universal ranking. It retains every B-versus-A reversal, reports all price and price-tree outcomes, and gives the exact interval-intersection rule with cumulative-work accounting. A common-refinement result proves that intersecting complete covers and taking the largest applicable leaf bound weakly dominates a scalar intersection, while disclosing its potentially multiplicative cell count. Only the scalar portfolio is claimed as executed. For the new price LP itself, the constant-price family is a nested restriction, so the exact restart-price LP has a genuine weak-dominance relation to that restriction. This is not a dominance theorem over aggregate products or over finite-time numerical proposals. The component ablations and an empirically validated adaptive scheduler remain separate unperformed experiments.

2.4. *R44-F4: Actual contribution of branching*

The original Q1 progress-by-time and progress-by-node curves remain in the article. The price-tree tables now show every root and final width, node count, and stopping reason. M0 improves from 0.003537 to 0.000331 in nine nodes and Q1 from 0.006410 to 0.000954 in 125 nodes. Those are complete-cover improvements that cross the fixed target. Conversely M3 only processes 23 price nodes and T3 remains open at 2,047 nodes. These failures remain in the main discussion. The revision does not infer practical effectiveness from eventual completeness and does not claim a strong-branching or reliability-branching comparison that was not run.

2.5. *R44-F5: SCIP lower bounds and the verification boundary*

SCIP native lower bounds, internally certified lower bounds, and independently checked upper policies are separate quantities throughout the article and data tables. A new restart-price lower bound is reconstructed independently from rational prices and Bellman inequalities; it is not a certification of SCIP's native global lower value. No native SCIP target is counted among the nine certified portfolio targets. The eight original native numerical targets remain visible as evidence favorable to the external solver. Exact post-certification of its full nonlinear search is not delivered in this revision. This objection therefore remains substantively open, rather than being declared solved through a shared interval or an unrelated exact-MILP feature.

2.6. *R44-F6: Complexity and finite precision*

The preserved complete-search theorem now has an explicit stopping modulus, a numerical floor, and a separate two-grid statement for the implemented probability rounding. These formulae display dependence on horizon, discount, action count, cost span, probability-box width, and arithmetic precision, including the exponential sufficient full-tree depth. The new price certificate has an explicit sparse LP representation size and can be replayed by one directed Bellman pass without solver duals. All price-tree proof sizes, peak construction memory, LP times, and node counts are tabulated; every independent replay has its own recorded time. We do not replace an exponential guarantee with a polynomial claim. The T3 run processes 2,047 nodes and still fails; that is retained, not selected away.

2.7. *R44-F7: Absolute success and cost units*

Absolute width 10^{-3} remains the primary criterion in every generated row and Boolean field. Relative width is shown only as a secondary diagnostic. The exact rational endpoints determine

target attainment, not rounded table values. The model costs are normalized departure charges, not monetary welfare estimates; an absolute width has meaning only in those declared units. The article gives the exact scaling rule: multiplying all costs by $c > 0$ multiplies the optimum and certified endpoints by c , so a comparable accuracy request must scale its absolute target by c . The historical target was fixed before observation and has not been retuned.

2.8. *R44-F8: Operating-optimal faces and exact ties*

The main theorem is now the restart-price certificate and its exact gap decomposition. It uses $0 \leq D \leq \varepsilon$ rather than a positive minimum action gap. Its Bellman minimum ranges over every action, including all operating-optimal face actions. At zero allowance a separate Bellman minimization over the whole zero-disadvantage face solves implementation cost exactly. The finite face dimension is reported for each model. In T0 it is 32, and the new restart certificate has width 3.0701×10^{-11} without branching. T1 and T2 also meet the target, whereas T3 remains open. These results directly strengthen the weak generic fallback while preserving the strict-gap quadratic theorem as a conditional sharper result.

2.9. *R44-F9: What the scaling experiments establish*

The three-family seven-allowance sequence is retained in full, with numerical theorem constants, arithmetic losses, and descriptive slopes separated from an exact-LP asymptotic statement. The sixteen price cases cover the original size and horizon sequence, and the four tie variants explicitly test degeneracy. The new 64-object precision experiment varies operating-witness accuracy and reports measured lower/upper losses next to analytic budgets. These are complementary diagnostics, not a full factorial study in n, T, m, β , transition sensitivity, and cost heterogeneity. No uniform empirical complexity law is fitted or claimed.

2.10. *R44-F10: Approximate operating witnesses*

The article adds a witness-stability proposition for restart prices and an upper-witness repair theorem. The lower error is bounded by the discounted sum of price times operating-enclosure width, plus directed evaluation error. The upper repair chooses the best action against the actual repaired continuation, with an explicit sufficient slack and cost-change bound. The executed audit constructs operating sub- and supersolutions at 16, 24, 32, and 40 bits, uses their conservative disadvantage in the lower endpoint, and checks the repaired policy against the upper witness at every restart. All 64 paired objects pass a reader that imports neither constructor nor optimizer. The sufficient guard fails in three 16-bit rows, where actual row checks still certify feasibility; no unsupported uniform upper bound is reported. This is a genuine approximate-witness interface test, but the witnesses are cheap finite directed Bellman enclosures. The requested difficult high-dimensional operating-oracle application remains unexecuted.

2.11. *R44-F11: Economic interpretation and application scope*

The model, installed policy, operating allowance, initial law, and implementation cost are defined before the numerical method. The supplement states the inventory and queue transitions and payoffs, including demand shocks, shortages, holding or service costs, departure charges, and terminal rewards. The article retains the distinction between all-restart protection and the initial-law objective, and the institutional assumption that state-contingent lotteries are admissible. Fixed lottery governance or communication costs are not silently set equal to observed data. All

three families remain designed known models with normalized units. No estimated application, sampling uncertainty, or externally calibrated welfare claim is manufactured in response to this request. The new certification theorems and executed tie improvements strengthen the methods contribution; an independent economic application remains necessary evidence beyond this package.

2.12. *R44-F12: The original controlled continuous-state problem*

The continuous-state target and all original results are retained. The restart-price theorem is proved for bounded Borel controlled kernels, not only exogenous fibers, and therefore allows action-dependent moving atoms. The preserved repair-density theorem gives feasible upper approximation by simple policies without smoothing those atoms. The price theorem adds a general lower certificate and an exact gap identity for the same common-policy object. These are substantive analytic advances on the motivating class, but they do not prove or execute a matched two-sided numerical convergence scheme for that class. All thirty positive-cost original randomized intervals and the inherited nonlinear intervals remain open and are stated as such in the abstract, results, and conclusion.

2.13. *R44-F13: Theorem-to-code contract*

Every new theorem appears with a proof in the main article. The accompanying contract matrix distinguishes mathematical assumptions, finite runtime checks, and unexecuted extensions. The restart lower reader verifies the finite Bellman recursion, the price-tree reader verifies the complete cover and price cuts, and the new witness reader checks both operating enclosures and the deployed common policy. The exact gap decomposition is recorded for all sixteen price policies. All 64 inherited R46 proof objects are independently replayed in the current revision. Finite tests are not described as machine-checked proofs of Borel measurability or continuum approximation.

2.14. *R44-F14: Ordinary source and final publication identity*

The final review tree exposes ordinary TeX, Python, JSON, CSV, and PDF files. The build creates a SHA-256 publication manifest, retains build logs, and records the source commit and inherited baseline. Any temporary transfer archive is a delivery mechanism rather than the scientific interface: its restored files are checked before compilation and committed normally. Inherited source transport and failed historical executions are preserved as historical files, not required to read the new article. No old branch is rewritten and no incomplete source chunk is treated as a final manuscript.

2.15. *R44-F15: Precise contribution relative to classical ingredients*

The article identifies its central mathematical result as the restart-price common-policy lower certificate with a nonnegative exact policy-to-bound gap decomposition. The decisive correction is the successor-price term involving the minimum of origin and successor prices; its residual is the explicitly priced shared-continuation mismatch. The proof permits controlled atoms and ties. Witness stability and continuation-best repair attach an explicit paired oracle budget. McCormick envelopes, RLT identities, linear optimization, and covering search remain named classical ingredients, not priority claims. A reproducible advantage is demonstrated on the exact and near-tie rows and I3, while all contrary B/A/SCIP results remain visible. This is not an assertion of uniform solver dominance or a comprehensive literature-priority theorem.

2.16. *R44-F16: Coherent article without loss of scientific content*

The main argument now follows one object: the minimum implementation cost of a common Markov policy protected at every restart. Its sequence is economic target, restart-price certificate and gap, complete finite search and repair, operating-witness precision, controlled-state approximation, and the complete numerical evidence. Structural aggregation and fiber identities retain their exact assumptions. The earlier full draft and the complete historical volume are preserved separately; technical details and adverse outcomes remain accessible rather than being deleted to create an appearance of coherence. The revision does not reduce the policy class or substitute an easier objective for the motivating problem.

3. ADDITIONAL TECHNICAL AND PRESENTATION COMMENTS

3.1. *Comment 1: Complete per-method reporting*

The supplement and CSV retain absolute/relative widths, target, stop, node count, LP/search time, proof bytes, and recorded construction peak memory. Independent replay times are separate. Unrecorded fine-grained memory or time components are marked unavailable rather than inferred.

3.2. *Comment 2: Fixed success criterion*

All target fields use the exact absolute width and the unchanged rational target 1/1000.

3.3. *Comment 3: Three different SCIP quantities*

Native SCIP lower, rational internal lower, and evaluated upper remain distinct in the preserved original tables and machine records. The new price bound is separately named.

3.4. *Comment 4: What verified means*

A verified SCIP candidate denotes only its feasible policy and implementation cost. Native global lower bounds remain numerical diagnostics.

3.5. *Comment 5: Portfolio intersection*

The displayed portfolio uses the maximum valid lower and minimum feasible upper for the identical model, policy class, and allowance. It closes nine of sixteen cases and charges all contributing work.

3.6. *Comment 6: Support-price grid*

The inherited fixed powers-of-two support grid is retained as an untuned design. The new restart-price LP optimizes nonnegative state/date prices rather than retuning that grid. No separate grid-sensitivity experiment is claimed.

3.7. *Comment 7: Strengthening ablations*

All constant/restart/root/search results are retained, but they do not isolate every support, cone, RLT, and local-proposal component. That full ablation is not reported as completed.

3.8. *Comment 8: Branching alternatives*

The actual scores and periodic widest-coordinate step remain in code and the finite stopping argument. Strong, pseudo-cost, and reliability branching have not been benchmarked.

3.9. *Comment 9: Unavailable LP proposals*

The original root status table explicitly reports missing primal/dual availability. Sound lower-corner or midpoint fallbacks remain in the code and proof cover. A counterfactual decomposition of their performance loss is not invented.

3.10. *Comment 10: Time components*

LP time, total construction/search time, and independent replay time are distinct. Serialization and every rational suboperation are not separately profiled in the inherited records.

3.11. *Comment 11: Proof size and nodes*

Complete proof-byte and node counts accompany both root and full price searches, retaining large-proof small-node cases. A descriptive proof-size plot is supplementary, not a scaling theorem.

3.12. *Comment 12: Verification memory*

No comparative peak-memory claim is made: inherited construction high-water measurements do not supply independently measured replay peaks.

3.13. *Comment 13: Bound histories*

All retained traces and exact covering proofs remain available. The revision does not fabricate ten improvements in cases where the trace contains fewer actual improvements.

3.14. *Comment 14: M3 root failure*

The original Bellman root has 17,952 variables versus 8,512 for aggregate and lacks a usable primal/dual proposal at its allocation. This is an observed formulation-size/status contrast, not proof that size alone caused the failure.

3.15. *Comment 15: Q3 formulation reversal*

Both formulations remain sound but neither finite relaxation contains the other as implemented. Q3 aggregate closes at the root while Bellman does not. The price development also closes Q3, without rewriting this adverse comparison.

3.16. *Comment 16: Optimal-action face*

Face dimensions are tabulated. Zero-allowance face optimization and the positive-allowance restart-price bound use all optimal actions. T0 has dimension 32 and is certified without branching.

3.17. *Comment 17: Malformed covers*

The inherited tests include omitted siblings/leaves, a coverage gap, and a cycle. The full reader checks unique/reachable cover nodes. New witness mutations attack a different proof interface; they do not claim exhaustive random-cover coverage.

3.18. *Comment 18: Separate adversarial targets*

Fifty-one new invalid witness objects cover endpoint, probability, operating enclosure, source/model/protocol identity, target, budget, and precision mutations, with three valid controls. They complement rather than replace inherited tree and price tests; exhaustive RLT-direction fuzzing remains unperformed.

3.19. *Comment 19: Per-invocation identity*

Every new witness invocation and each R46 price/tree invocation checks the model and protocol against their frozen identities. The current replay checks all 64 inherited R46 objects.

3.20. *Comment 20: Zero fallback*

Nonnegative cost gives a valid zero lower bound. Zero lower bounds and zero-cost exact rows are not evidence that a difficult positive optimum has been identified.

3.21. *Comment 21: Economic primitives*

The supplement writes the inventory and queue payoff and transition equations, installed actions, departure charges, shocks, and terminal rewards. They remain designed examples.

3.22. *Comment 22: Uniform initial laws*

The frozen uniform laws are stated as experimental choices, not empirically justified distributions. They are not changed after observation.

3.23. *Comment 23: Nonuniform and atomic laws*

The theorem permits arbitrary initial laws, including point masses. A new prospective sensitivity suite over initial laws has not been executed.

3.24. *Comment 24: Multiple operating constraints*

The lower theorem extends componentwise in prices and allowances. Upper repair additionally requires a common feasible reference/slack; separately optimal references need not coincide. No general multi-constraint repair is asserted without that condition.

3.25. *Comment 25: Cost-scale sensitivity*

Costs, objective, certificates, and absolute accuracy scale together under multiplication by a positive constant. Normalized historical units and the fixed original target are retained.

3.26. *Comment 26: Local proposal budget*

The original local proposal is called inside the method clock; the clock is checked between complete operations and may overrun the nominal limit. R46 price-search local improvement is disabled and inherited incumbent construction is excluded from incremental time, explicitly disclosed.

3.27. *Comment 27: No opaque review interface*

Ordinary restored sources and compiled documents form the final review object. Earlier transport records remain historical only.

3.28. *Comment 28: Prior R43 response*

The full R45 response to all seventeen R43 findings and thirty additional comments is preserved in the historical source and reading volume. This current response separately covers all sixteen R44 findings and thirty comments.

3.29. *Comment 29: Original continuous intervals*

The abstract explicitly states that all thirty original positive-cost randomized intervals remain open.

3.30. *Comment 30: Three levels of conclusion*

The conclusion separates analytic validity, finite arithmetic replay, and observed target-level closure. It also distinguishes an asymptotic approximation statement from an executed two-sided continuum solver.

4. READING AND VERIFICATION MAP

The main article presents the economic target, all general theorem statements, and their proofs. The supplement retains every original and new numerical row. The computation report identifies exact inputs, reader boundaries, commands, and hashes. The historical volume contains the full R45 draft and prior historical material, including the R43 response. The publication manifest binds ordinary sources, generated tables, and PDFs. A passing finite reader is not a machine-checked general theorem; a reported native solver bound is not a certified lower endpoint; and a retrospective development is not a new prospective experiment.